

13.6 Classifying Triangles

Lesson Introduction

 Benchmark	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.		 Learning Targets	- I can use coordinate geometry to classify triangles in a mathematical or real-world context. - I can draw a triangle on a coordinate plane using given definitions, properties, or theorems.
 Benchmark Coherence	Building On MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through a given point.		Working Toward N/A	
 Essential Questions	- How can coordinate geometry be used to classify triangles on the coordinate plane? - How can the Pythagorean Theorem be used to find the distance between two points on the coordinate plane? - What are the characteristics of equilateral, isosceles, and scalene triangles?			
 Lesson Narrative	Lesson 13.6 guides students through the process of using the Pythagorean Theorem to calculate distance on the coordinate plane, building from content knowledge from 8 th grade. The lesson requires students to graph and use the distance formula and/or Pythagorean Theorem to find the measures of lengths to identify types of triangles. Students are expected to recall terms such as equilateral, scalene, and isosceles, and identify their characteristics.			
 Component Summary	13.6.1 Warm-Up (~5 mins.)	Notice and Wonder (SE p. 315)	13.6.3 Exploration (15 mins.)	Identifying types of triangles on coordinate plane (SE p. 316)
	13.6.2 Refresher (5 mins.)	Refresher on using the Pythagorean Theorem (SE p. 315)	13.6.4 Your Turn! (20 mins.)	Practice on identifying triangles on the coordinate plane (SE p. 319)
	13.6.5 Wrap-Up (~5 mins)	Students summarize their learning (SE p. 321)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Equilateral Triangle - Isosceles Triangle - Scalene Triangle		- Calculators (Optional) Graphing Utility (Optional) Colored pencils/pens	
	<u>Additional Preparation</u> If necessary, prepare a graphing utility, such as Desmos or GeoGebra.			

 Teacher Tips	Common Misconceptions - Students may incorrectly set up the Pythagorean Theorem, and may forget to find the square root of c^2. - Students may incorrectly identify whether a triangle is equilateral, isosceles, or scalene. - Students may incorrectly use the Pythagorean Theorem to find distance.	Things to Consider - Consider allowing students who need extra guidance and support to use a graphing utility, such as GeoGebra, to help students see the relationships between the triangles. - Consider allowing students to use colored pencils/pens during this lesson to label the sides of the triangles. - Determine if your students need the support of all or part of the Refresher. - In problems that require several steps, students should not round until the final step. Rounding mid-step could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common careless errors that many students make.
	Literacy and Language Routines Anticipate, Monitor, Select, Sequence, Connect The five practices for orchestrating mathematical discussion can be implemented effectively in Lesson 13.6. Anticipate the different strategies and interpretations and pay attention to students' thinking processes. Select which students can share their solutions, and sequence by common misconception or strategy. Finally, help students connect their thinking and notice patterns of different solutions and processes.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.3.1: Mathematical Fluency Students will have to select efficient and appropriate methods for solving problems. This will require students to maintain flexibility and accuracy, while performing the procedures necessary to solve the problems. Students will have to adapt their knowledge of the Pythagorean Theorem to apply it to new contexts.
	Consider differentiating the problems in the Your Turn! section of the lesson. Give each problem a point value that is equivalent to the number. For example, question 1 is worth 1 point, question 2 is worth 2 points, and so on. Students are required to earn a minimum of five points. Structuring the Your Turn! using this format gives students an opportunity to have choice and will require students to do a mixture of lower and higher-level problems.	
 Differentiate Instruction	Lesson Note	

13.6.1 Warm-Up: Latitude and Longitude

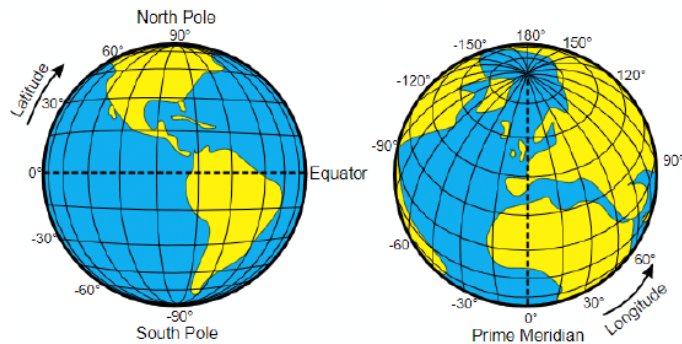
Component Overview: Students read the prompt about latitude and longitude and are given a table of the latitude and longitude of several cities in Florida. Students are asked to notice and wonder.



13.6.1 Warm-Up: Latitude and Longitude

In ancient times, people used landmarks to identify locations, but travelers were not familiar with local landmarks, making traveling difficult. Hipparchus, a Greek astronomer, was the first to specify location using latitude and longitude as coordinates. Hipparchus applied his knowledge of spherical angles to the problem of denoting locations on the Earth's surface.

Latitude is the measurement of the globe north to south, and longitude is the measurement of the globe east to west.



The table shows the latitude and longitude of different Florida cities.

City	Coordinates	City	Coordinates
Panama City	30.2N, 85.7W	Miami	25.8N, 80.2W
Tallahassee	30.46N, 84.28W	Tampa	28.0N, 82.5W
Lake Worth	26.6N, 80.1W	Orlando	28.5N, 81.4W
Jacksonville	30.3N, 81.7W	Gainesville	29.7N, 82.3W

1. What do you notice? What do you wonder?

Answers will vary. The latitude values are all close to 30, and the longitude values are near 80. How far apart are the cities in miles from each other?



Give students an opportunity to share their Notice and Wonder and see if there are any commonalities.



Consider having students take an educated guess on the longitude and latitude of your school, using the information given.

13.6.2 Refresher: Finding Distance

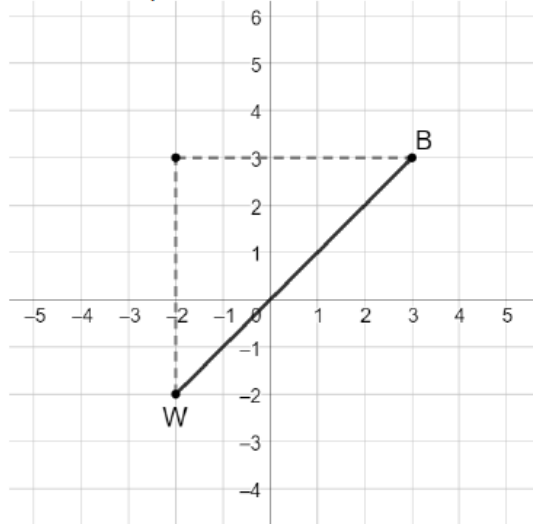
Component Overview: Students review how to use the Pythagorean Theorem to find the distance on the coordinate plane.



13.6.2 Refresher: Finding Distance

Work with your partner to complete the following.

- Find the length of $\overline{WB}$ using the Pythagorean Theorem. If necessary, round to the nearest thousandth.



$$\begin{aligned} 5^2 + 5^2 &= WB^2 \\ 25 + 25 &= WB^2 \\ 50 &= WB^2 \\ \sqrt{50} &= WB \\ 7.071 &= WB \end{aligned}$$

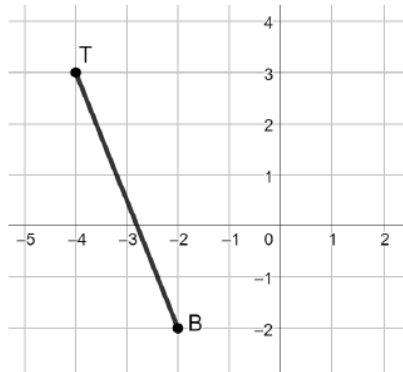


Begin by asking students what they remember about the Pythagorean Theorem, which is used extensively in Unit 7, as well as other units throughout the course. Consider using the following questions:

- What is the Pythagorean Theorem used for?
- How can you use the Pythagorean Theorem to find the length of a side of a right triangle?

13.6.2 Refresher: Finding Distance (continued)

2. Use the Pythagorean Theorem to find the length of $\overline{TB}$. If necessary, round to the nearest thousandth.



$$\begin{aligned} 5^2 + 2^2 &= TB^2 \\ 25 + 4 &= TB^2 \\ 29 &= TB^2 \\ \sqrt{29} &= \sqrt{TB^2} \\ 5.385 &= TB \end{aligned}$$

3. Explain how the Pythagorean Theorem can be used to find the distance between two points.

Answers will vary. Find the horizontal and vertical distance between the two endpoints. This can be done by using the gridlines on the coordinate grid or by finding the difference between the x-values and the difference between the y-values of the endpoints. Then, substitute the horizontal distance and vertical distance between the points for a and b in the Pythagorean Theorem, $a^2 + b^2 = c^2$. Solve to find the length of the line segment, which will be the hypotenuse of the right triangle.



Consider opportunities to provide differentiation using a visual approach. Have students draw a right triangle on the coordinate plane on question 2 to help better visualize how the Pythagorean Theorem can be used to find distance on the coordinate plane.



Tie question 3 to the MLR of Anticipate, Monitor, Select, Sequence, and Connect. Consider the ways that students will answer this question and select a couple of responses to discuss as a lead into using the Pythagorean Theorem to find distance and length on the coordinate plane.

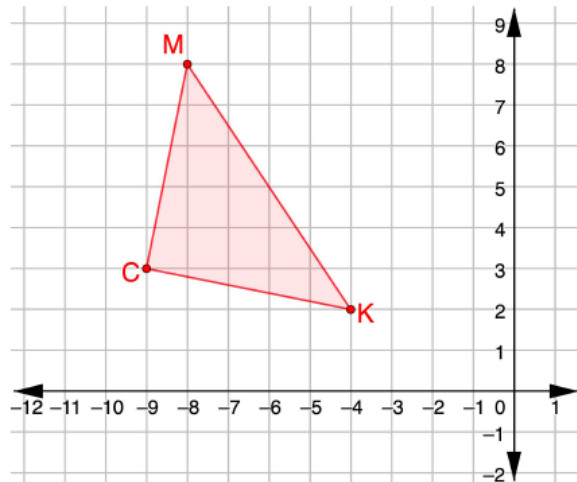
13.6.3 Exploration: Classifying Triangles

Component Overview: This Exploration has students graph triangles on the coordinate plane to find the distances of each side. Students use the distances to classify the triangles as equilateral, isosceles, and scalene, based on the number of congruent sides.



13.6.3 Exploration: Classifying Triangles

1. $\triangle MKC$ has vertices at $M(-8,8)$, $K(-4,2)$, and $C(-9,3)$.
 - a. Graph $\triangle MKC$ on the coordinate plane.



Consider allowing students to use a tool such as GeoGebra to graph the triangles. GeoGebra can be used to help visualize the relationship between the values of the segments and the type of triangle that is formed.

- b. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Length of Line Segment
$\overline{CM}$	5.099
$\overline{MK}$	7.211
$\overline{KC}$	5.099

- c. What do you notice about the side lengths of $\triangle MKC$?
Two sides have the same lengths.

13.6.3 Exploration: Classifying Triangles (continued)

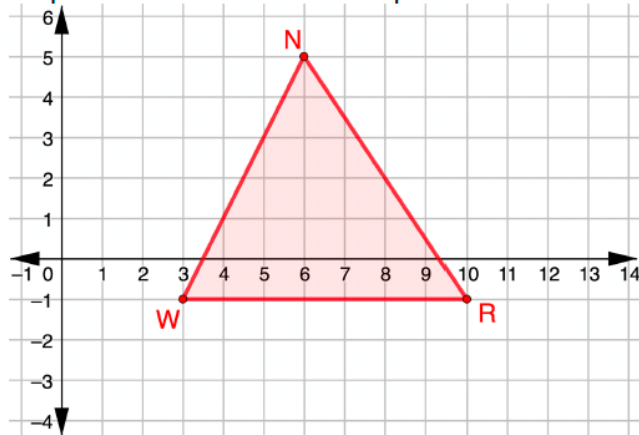
d. Complete the statement.

$\triangle MKC$ is classified as a(n)
☐ equilateral
☒ isosceles
☐ scalene
 triangle

because $\triangle MKC$ has
☐ no
☒ two
☐ three
 equal side lengths.

2. $\triangle NWR$ has vertices at $N(6,5)$, $W(3,-1)$, and $R(10,-1)$.

a. Graph $\triangle NWR$ on the coordinate plane.



b. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Length of Line Segment
$\overline{NR}$	7.211
$\overline{RW}$	7
$\overline{WN}$	6.708



Have students make a prediction about which type of triangle is on the coordinate plane before students complete the table. Doing this will tie into the Literacy and Language Routine by having students looking at several different responses and reasonings.

Be careful that students do not rely on their prediction and choose to not support it mathematically.

13.6.3 Exploration: Classifying Triangles (continued)

- c. What do you notice about the side lengths of $\triangle NWR$?

None of the sides have the same length.

- d. Complete the statement.

$\triangle NWR$ is classified as a(n)
☐ equilateral
☐ isosceles
☒ scalene
 triangle

because $\triangle NWR$ has
☒ no
☐ two
☐ three
 equal side lengths.



Consider polling students for their prediction on what type of triangle $\triangle NWR$ is. Because the measures of the side lengths are close, the triangle does not appear to be scalene. It may be helpful for students to understand why their predictions may be incorrect.

3. $\triangle MSG$ has vertices at $M(-3, 4)$, $S(2.2, 1)$, and $G(-3, -2)$.

- a. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Length of Line Segment
$\overline{MS}$	6.003
$\overline{SG}$	6.003
$\overline{GM}$	6

- b. Complete the statement.

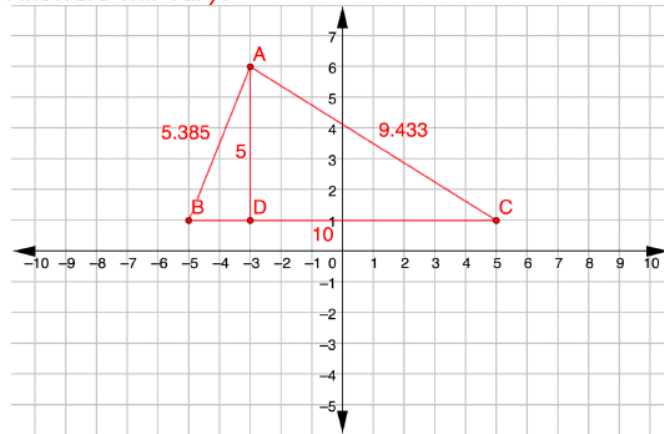
$\triangle MSG$ is classified as a(n)
☐ equilateral
☒ isosceles
☐ scalene
 triangle

because $\triangle MSG$ has
☐ no
☒ two
☐ three
 equal side lengths.

13.6.3 Exploration: Classifying Triangles (continued)

4. Draw a scalene triangle with one vertex at $(-3, 6)$ and with a base that is twice the length of its height.

Answers will vary.



Use the following questions to help students construct their triangle for question 4:

- What step would be completed first to begin constructing the triangle?
- How many units do you want to draw the height of the triangle?
- How can you find the length of the base?



Have students compare their triangles with a partner or another student in the classroom. If time allows, consider sharing a few examples with the class.

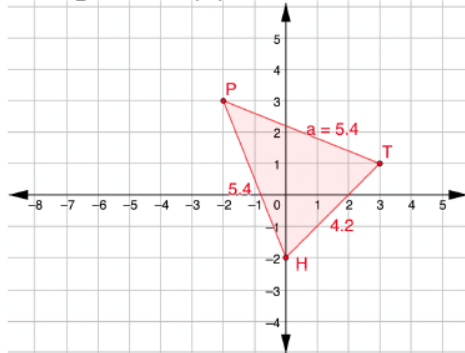
13.6.4 Your Turn!

Component Overview: Students work independently to practice classifying triangles on the coordinate plane.



13.6.4 Your Turn!

1. $\triangle PTH$ has vertices at $P(-2,3)$, $T(3,1)$, and $H(0,-2)$. Determine if $\triangle PTH$ is an equilateral, isosceles, or scalene triangle. Justify your answer with math.



$$\begin{aligned} 3^2 + 3^2 &= HT^2 \\ 9 + 9 &= HT^2 \\ 18 &= HT^2 \\ \sqrt{18} &= \sqrt{HT^2} \\ 4.243 &\approx HT \end{aligned}$$

$$\begin{aligned} 2^2 + 5^2 &= PT^2 \\ 4 + 25 &= PT^2 \\ 29 &= PT^2 \\ \sqrt{29} &= \sqrt{PT^2} \\ 5.385 &\approx PT \end{aligned}$$

$$\begin{aligned} 2^2 + 5^2 &= PH^2 \\ 4 + 25 &= PH^2 \\ 29 &= PH^2 \\ \sqrt{29} &= \sqrt{PH^2} \\ 5.385 &\approx PH \end{aligned}$$

PH and PT are the same length. This is an isosceles triangle.



Consider having students work in groups of three. Each student in the group is responsible for finding the length of one side of the triangle. Together as a group, they will review each other's work, and ask clarifying questions, if needed. Then, the group will determine the type of triangle that is given.

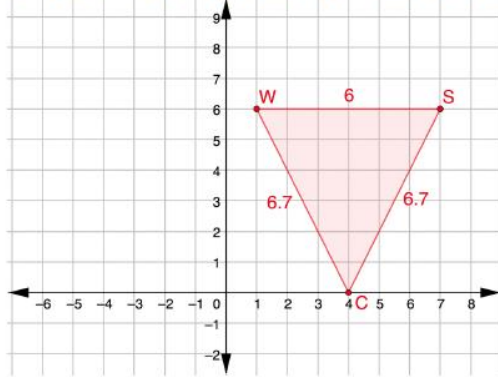


Consider having students create a table to organize their work for question 1.

13.6.4 Your Turn! (continued)

2. Given: $\triangle SCW$ with $W(1,6)$, $S(7,6)$, and $C(4,0)$

a. Show: $\triangle SCW$ is **NOT** an equilateral triangle



$$\begin{aligned} 3^2 + 6^2 &= CS^2 \\ 9 + 36 &= CS^2 \\ 45 &= CS^2 \\ \sqrt{45} &= \sqrt{CS^2} \end{aligned}$$

$$6.708 \approx CS$$

$$\begin{aligned} 3^2 + 6^2 &= CW^2 \\ 9 + 36 &= CW^2 \\ 45 &= CW^2 \\ \sqrt{45} &= \sqrt{CW^2} \\ 6.708 &\approx CW \end{aligned}$$

$$WS = 6$$

The triangle is not equilateral because all of the sides are not the same.

b. What type of triangle is $\triangle SCW$?

It is an isosceles triangle because two of the sides have the same length.

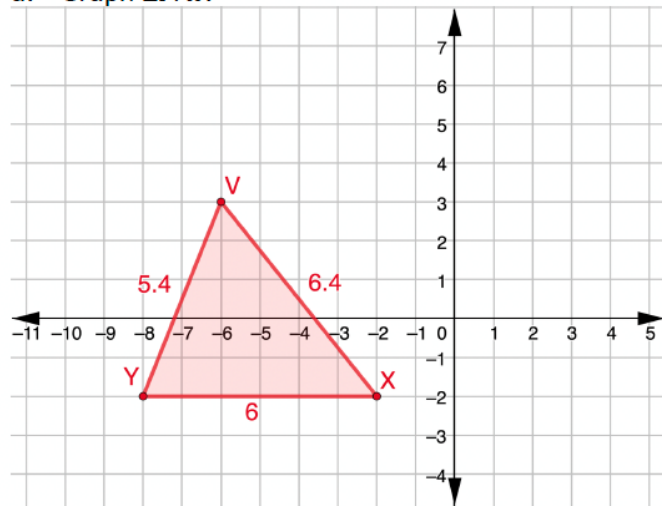


Students may be tempted to not show work for this problem. Recommend that students show work on question 2.

13.6.4 Your Turn! (continued)

3. Diego is creating a triangular garden and uses the $\triangle YVX$ with vertices at $Y(-8, -2)$, $V(-6, 3)$, and $X(-2, -2)$ to design the garden.

a. Graph $\triangle YVX$.



- b. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Side Length
$\overline{YV}$	5.385
$\overline{VX}$	6.403
$\overline{XY}$	6

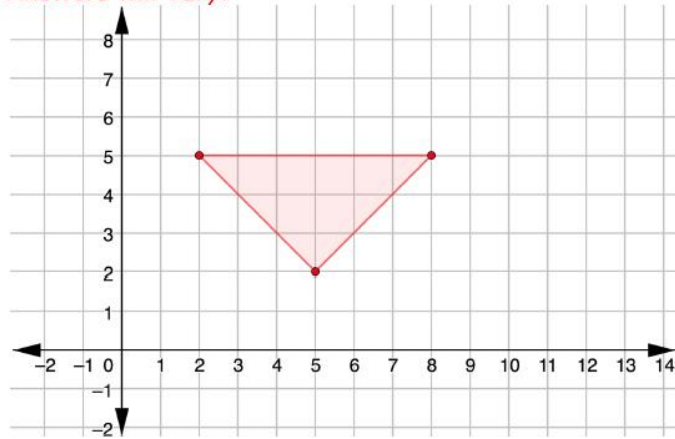
- c. Classify the type of triangular garden Diego is creating.

It is a scalene triangular garden.

13.6.4 Your Turn! (continued)

4. Draw an isosceles triangle whose base is three more than the height of the triangle and has one vertex at $(2,5)$.

Answers will vary.



Use Anticipate, Monitor, Select, Sequence, and Connect for question 4, and have students share their triangles to see if students created different triangles.



Consider allowing students to try and create their triangle in GeoGebra as a safe place, before drawing it on the coordinate plane provided.



For students who finish early, challenge them to create their own triangles, and have other students identify whether the triangle is equilateral, isosceles, or scalene using coordinate geometry.

13.6.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



13.6.5 Wrap-Up: Cool Down

1. Identify **two** strategies you can use to classify triangles using coordinate geometry.

Answers will vary.



Consider the online resources for additional enrichment opportunities for students to practice or test their abilities.